Elementary Analysis

Chapter 3, Section 18

Properties of Continuous Functions
Exercises 18.1–18.12, with every subpart

Mathjock

Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Chapter 3. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

Contents

18 Properties of continuous functions **2**

18.1 Exercise 18.1: A maximum after reflection is a minimum before reflection 2
18.2 Exercise 18.2: Where the closed-interval proof uses the endpoints 2
18.3 Exercise 18.3: A polynomial on a half-open interval 3
18.4 Exercise 18.4: A missing limit point produces an unbounded continuous function . . 4
18.5 Exercise 18.5: Intersecting graphs and a fixed point 5
18.6 Exercise 18.6: A fixed point of cosine inside a specified interval 6
18.7 Exercise 18.7: An exponential equation with an interior solution 6
18.8 Exercise 18.8: Opposite signs force a real zero 6
18.9 Exercise 18.9: Every polynomial of odd degree has a real root 7
18.10 Exercise 18.10: Equal values exactly one unit apart 8
18.11 Exercise 18.11: The decreasing versions of the inverse-function theorems 9
18.12 Exercise 18.12: The intermediate value property without continuity 10

18 Properties of continuous functions

The extreme value theorem concerns a continuous function on a closed, bounded interval. The intermediate value theorem concerns a continuous function on any interval. These conclusions answer different questions: the first guarantees attained extreme values, while the second fills in values between two already attained values.

All continuity statements below are relative to the specified domain. For example, an inverse defined on a half-closed interval is tested only at points belonging to that interval. Multiplying an inequality by a negative number reverses its direction; we will use that fact explicitly.

18.1 Exercise 18.1: A maximum after reflection is a minimum before reflection

What we must show. Suppose f is continuous on $[a, b]$ and $-f$ assumes its maximum at $x_0 \in [a, b]$. Show that f assumes its minimum there.

Translate the meaning of maximum into an inequality.

For every $x \in [a, b]$, the maximum assumption means

$$(-f)(x) \leq (-f)(x_0), \qquad \text{that is,} \qquad -f(x) \leq -f(x_0).$$

Multiplying both sides by -1 reverses the order and gives $f(x) \geq f(x_0)$. Equivalently,

$$f(x_0) \leq f(x) \qquad \text{for every } x \in [a, b].$$

Because x_0 belongs to the domain, the value $f(x_0)$ is actually attained, rather than merely being a lower bound.

Conclusion. The displayed inequality is exactly the definition of a minimum of f at x_0. Continuity is used by the theorem to obtain the maximum of $-f$; this final order reversal needs no additional continuity argument.

18.2 Exercise 18.2: Where the closed-interval proof uses the endpoints

What we must show. Locate the failure in the proof of Theorem 18.1 when its domain $[a, b]$ is replaced by (a, b), and explain the consequences.

1. A bounded input sequence still has a convergent subsequence.

Every sequence (x_n) in (a, b) is bounded, so Bolzano–Weierstrass still provides a subsequence x_{n_k} converging to a real number x_0. The difficulty is its location. The inequalities $a < x_{n_k} < b$ imply only $a \leq x_0 \leq b$. They do not imply the strict inequalities $a < x_0 < b$. Thus the subsequence may tend to an omitted endpoint.

2. Identify the missing step in the boundedness argument.

In the theorem's contradiction argument, the selected inputs satisfy $|f(x_n)| > n$. With a closed interval, the subsequential limit x_0 lies in the domain, so continuity gives $f(x_{n_k}) \to f(x_0)$, a finite real number. On an open interval, x_0 may be outside the domain. Then $f(x_0)$ is not defined, and continuity of f on (a, b) gives no statement at x_0. The contradiction is lost precisely at that step.

For a concrete example, $f(x) = 1/x$ is continuous on $(0, 1)$. The sequence $x_n = 1/(n + 1)$ lies in that interval and tends to the excluded point 0, while $f(x_n) = n + 1$ is unbounded. There is no value $f(0)$ to which continuity could force these outputs to converge.

3. The same missing-domain issue affects attainment of the bounds.

Even if f is bounded, a sequence (y_n) chosen so that $f(y_n) \to \sup f((a, b))$ may have its input subsequence tend to an endpoint. One then cannot identify the limiting output as a value of f at an allowed input.

For example, $f(x) = x$ is continuous and bounded on $(0, 1)$, with infimum 0 and supremum 1. It has neither a minimum nor a maximum there. In fact, for every $x \in (0, 1)$,

$$0 < \frac{x}{2} < x < \frac{x + 1}{2} < 1.$$

The points $x/2$ and $(x + 1)/2$ give strictly smaller and strictly larger function values than $f(x)$. The sequences approaching the bounds tend to the excluded endpoints 0 and 1.

Conclusion. Bolzano–Weierstrass still works, but its limit need not belong to the open domain. Both the boundedness contradiction and the extremum-attainment step require continuity at that limit. An individual function on an open interval may still be bounded and attain extrema; those properties are no longer guaranteed.

18.3 Exercise 18.3: A polynomial on a half-open interval

What we must show. Use calculus to determine the extreme values of

$$f(x) = x^3 - 6x^2 + 9x + 1 \qquad (0 \le x < 5).$$

Plan. Find the stationary points, inspect the derivative's sign, and remember that 5 is not an allowed input.

1. Differentiate and find the interior stationary points.

The power rule gives

$$f'(x) = 3x^2 - 12x + 9 = 3(x^2 - 4x + 3) = 3(x - 1)(x - 3).$$

Thus the only zeros of f' are 1 and 3, both in the domain. Its signs on the intervening intervals are

x	$(0, 1)$	$(1, 3)$	$(3, 5)$
$x - 1$	$-$	$+$	$+$
$x - 3$	$-$	$-$	$+$
$f'(x)$	$+$	$-$	$+$

The mean value theorem therefore makes f increasing on $[0, 1]$, decreasing on $[1, 3]$, and increasing on $[3, 5)$.

2. Evaluate all the candidates and the omitted endpoint value.

Substitution gives

$$f(0) = 1,$$
$$f(1) = 1 - 6 + 9 + 1 = 5,$$
$$f(3) = 27 - 54 + 27 + 1 = 1,$$
$$f(5) = 125 - 150 + 45 + 1 = 21.$$

The final line evaluates the polynomial formula at 5 to find the endpoint limit. It does not add 5 to the stated domain.

3. Establish and locate the global minimum.

The useful factorization

$$f(x) = x(x-3)^2 + 1$$

shows directly that $f(x) \geq 1$ for every $x \in [0, 5)$: the factor x and the square $(x-3)^2$ are nonnegative. Equality requires $x(x-3)^2 = 0$, which holds exactly when $x = 0$ or $x = 3$. Both points are in the domain.

4. Explain why there is no global maximum.

On $[0, 3]$, the derivative analysis gives $f(x) \leq 5$. On $[3, 5)$, strict increase toward the omitted endpoint gives $f(x) < 21$. In addition, $x_n = 5 - 1/n$ lies in $[0, 5)$ and tends to 5, so continuity of the polynomial formula on $\mathbb{R}$ gives $f(x_n) \to 21$. Thus 21 is the supremum but is not attained.

One can also exclude a maximum without using the word supremum. For $x \in [3, 5)$, the point $y = (x + 5)/2$ satisfies $x < y < 5$, and strict increase gives $f(y) > f(x)$. For $x \in [0, 3]$, we have $f(x) \leq 5$, whereas $f(4) = 64 - 96 + 36 + 1 = 5$ and strict increase on $[3, 5)$ gives $f(9/2) > f(4) = 5$. Thus every allowed input has some allowed input with a larger function value.

Conclusion. The minimum is 1, attained exactly at $x = 0$ and $x = 3$. There is no maximum. The supremum is 21, approached as $x \to 5$ from the left. The value 5 at $x = 1$ is a local maximum, not a global maximum on $[0, 5)$.

18.4 Exercise 18.4: A missing limit point produces an unbounded continuous function

What we must show. Suppose $S \subseteq \mathbb{R}$ contains a sequence (x_n) converging to $x_0 \notin S$. Construct a continuous function on S whose values are unbounded.

1. Choose a function that grows near the excluded point.

Define

$$F(x) = \frac{1}{|x - x_0|} \qquad (x \in S).$$

Every $x \in S$ differs from x_0, so the denominator is positive and $F(x)$ is a well-defined finite real number. The function is a quotient of continuous functions with a nonzero denominator, restricted to S, and hence is continuous there.

2. Verify continuity directly, including the domain restriction.

Fix $y \in S$ and let $d = |y - x_0| > 0$. If $x \in S$ and $|x - y| < d/2$, the triangle inequality yields

$$|x - x_0| \geq |y - x_0| - |x - y| > d/2.$$

The reverse triangle inequality then gives

$$|F(x) - F(y)| = \frac{||y - x_0| - |x - x_0||}{|x - x_0|\,|y - x_0|} \leq \frac{|x - y|}{|x - x_0|\,d} \leq \frac{2\,|x - y|}{d^2}.$$

Given $\varepsilon > 0$, choose

$$\delta = \min\left\{\frac{d}{2}, \frac{\varepsilon d^2}{2}\right\} > 0.$$

For every $x \in S$ with $|x - y| < \delta$, the estimate is strictly less than ε. Thus F is continuous at every $y \in S$, in precisely the domain-relative sense.

3. Use the given sequence to disprove boundedness.

Let $M > 0$. Since $x_n \to x_0$, there is an integer N such that $n > N$ implies $|x_n - x_0| < 1/M$. All these distances are nonzero because $x_n \in S$ and $x_0 \notin S$. Reciprocating the positive inequality gives

$$F(x_n) = \frac{1}{|x_n - x_0|} > M \qquad (n > N).$$

Thus the function takes values larger than every proposed positive bound. In fact, $F(x_n) \to +\infty$.

Conclusion. The function $F(x) = 1/|x - x_0|$ is continuous on S and unbounded there. Its singular point lies outside its domain.

18.5 Exercise 18.5: Intersecting graphs and a fixed point

What we must show. Prove (a) that the stated endpoint inequalities force two continuous graphs to meet, and (b) that Example 1 follows as a special case.

(a), first: subtract the functions.

Assume f and g are continuous on $[a, b]$, with $f(a) \geq g(a)$ and $f(b) \leq g(b)$. Define $h(x) = f(x) - g(x)$ on this interval. The difference of continuous functions is continuous, and

$$h(a) \geq 0, \qquad h(b) \leq 0.$$

If $h(a) = 0$, take $x_0 = a$. If $h(b) = 0$, take $x_0 = b$. Either choice already gives $f(x_0) = g(x_0)$.

(a), second: use the strict sign change in the remaining case.

If neither endpoint value is zero, the weak inequalities become $h(a) > 0$ and $h(b) < 0$. Thus 0 is strictly between those values. The intermediate value theorem gives $x_0 \in (a, b)$ with $h(x_0) = 0$. By the definition of h, this equation is $f(x_0) - g(x_0) = 0$, or $f(x_0) = g(x_0)$. If the interval degenerates to $a = b$, the two given inequalities at that single point already force equality, so that case also is covered.

(b) Compare a map of the unit interval with the identity.

In Example 1, a continuous function f maps $[0, 1]$ into $[0, 1]$. Let $g(x) = x$ on $[0, 1]$, a continuous polynomial. The image restriction gives

$$f(0) \geq 0 = g(0), \qquad f(1) \leq 1 = g(1).$$

Part (a) therefore supplies $x_0 \in [0,1]$ with $f(x_0) = g(x_0) = x_0$. This is the fixed-point conclusion of the example, and it includes possible fixed points at 0 or 1.

Conclusion. The endpoint order reversal forces an intersection. Comparing a continuous map $[0,1] \to [0,1]$ with $g(x) = x$ recovers Example 1.

18.6 Exercise 18.6: A fixed point of cosine inside a specified interval

What we must show. Prove that $x = \cos x$ for at least one $x \in (0, \pi/2)$.

1. Move both sides to one continuous function.

Set $h(x) = x - \cos x$ on $[0, \pi/2]$. The identity and cosine are continuous, so their difference is continuous on this closed interval.

2. Compute opposite nonzero signs at the endpoints.

The familiar trigonometric values give

$$h(0) = 0 - 1 = -1 < 0, \qquad h(\pi/2) = \pi/2 - 0 = \pi/2 > 0.$$

Consequently the target value 0 is strictly between the endpoint values. By the intermediate value theorem there is $x \in (0, \pi/2)$ with $h(x) = 0$.

Conclusion. At this interior point, $h(x) = 0$ means $x - \cos x = 0$, which is the required equation $x = \cos x$.

18.7 Exercise 18.7: An exponential equation with an interior solution

What we must show. Prove that $xe^x = 2$ for some $x \in (0,1)$.

1. Form a continuous function whose zero solves the equation.

Let $h(x) = xe^x - 2$ on $[0,1]$. The identity and exponential are continuous, and the product and subtraction rules make h continuous as well.

2. Check the signs, including the inequality involving e.

At zero, $e^0 = 1$, so $h(0) = 0 \cdot 1 - 2 = -2 < 0$. At one, $h(1) = e - 2$. The identity $e = \sum_{k=0}^{\infty} 1/k!$, stated in Section 16, shows that

$$e \geq 1 + 1 + \frac{1}{2} = \frac{5}{2} > 2,$$

because the omitted terms are nonnegative. Hence $h(1) > 0$.

3. Apply the intermediate value theorem with strict endpoint signs.

There is $x \in (0,1)$ with $h(x) = 0$, since zero lies strictly between $h(0)$ and $h(1)$.

Conclusion. The resulting equation is $xe^x - 2 = 0$, so $xe^x = 2$ at an allowed interior point.

18.8 Exercise 18.8: Opposite signs force a real zero

What we must show. Suppose $f : \mathbb{R} \to \mathbb{R}$ is continuous and $f(a)f(b) < 0$. Prove that f has a zero strictly between a and b.

1. Extract the endpoint information from the product.

A negative product of two real numbers means one factor is strictly positive and the other strictly negative. In particular, neither factor is zero. Also $a \neq b$, since otherwise the product would be the square $f(a)^2 \geq 0$.

Put $\alpha = \min\{a, b\}$ and $\beta = \max\{a, b\}$. Then $\alpha < \beta$, and $f(\alpha)$ and $f(\beta)$ still are the two values of opposite signs. Their order may be either positive then negative, or negative then positive; in both cases zero lies strictly between them.

2. Restrict to the ordered interval and apply the theorem.

The restriction of f to $[\alpha, \beta]$ is continuous. The intermediate value theorem provides $c \in (\alpha, \beta)$ with $f(c) = 0$.

Conclusion. The point c is strictly between the original endpoints, regardless of which of a and b was larger.

18.9 Exercise 18.9: Every polynomial of odd degree has a real root

What we must show. For a polynomial

$$f(x) = a_0 + a_1 x + \cdots + a_m x^m, \qquad m \text{ odd}, \quad a_m \neq 0,$$

prove that $f(c) = 0$ for some real c.

Plan. At inputs of sufficiently large magnitude, the leading odd power dominates all the lower powers. Give an explicit bound rather than assume a limit theorem for functions from the later Section 20.

1. Normalize the leading coefficient without changing the zeros.

Since $a_m \neq 0$, define

$$g(x) = \frac{f(x)}{a_m} = x^m + \sum_{j=0}^{m-1} b_j x^j, \qquad b_j = \frac{a_j}{a_m}.$$

The functions f and g have exactly the same zeros: $g(x) = 0$ if and only if $f(x) = a_m g(x) = 0$. Both are polynomials and hence are continuous on $\mathbb{R}$.

2. Bound the lower-degree part at large positive and negative inputs.

Set

$$C = \sum_{j=0}^{m-1} |b_j| \geq 0, \qquad R = \max\{1, C\} + 1.$$

Thus $R > 1$ and $R > C$. For either $x = R$ or $x = -R$, we have $|x| = R$. Since $0 \leq j \leq m-1$ and $R \geq 1$, $R^j \leq R^{m-1}$. Applying the triangle inequality gives

$$\left| \sum_{j=0}^{m-1} b_j x^j \right| \leq \sum_{j=0}^{m-1} |b_j| \, R^j \leq C R^{m-1}.$$

3. Use the odd parity of m to obtain opposite signs.

At R, the preceding estimate gives

$$g(R) \geq R^m - CR^{m-1} = R^{m-1}(R - C) > 0.$$

At $-R$, the leading term is $(-R)^m = -R^m$ because m is odd, so

$$g(-R) \leq -R^m + CR^{m-1} = -R^{m-1}(R - C) < 0.$$

Both strict signs follow from $R > 0$ and $R - C > 0$. The argument includes $m = 1$: the lower-degree part then is just the constant b_0.

4. Fill in zero between those two values.

Continuity of g on $[-R, R]$ and its opposite endpoint signs give a point $c \in (-R, R)$ with $g(c) = 0$ by the intermediate value theorem. Hence $f(c) = a_m g(c) = 0$.

Conclusion. Every real polynomial of odd degree has a real root. The proof treats every nonzero leading coefficient, including a negative one, through the normalization $g = f/a_m$.

18.10 Exercise 18.10: Equal values exactly one unit apart

What we must show. Suppose f is continuous on $[0, 2]$ and $f(0) = f(2)$. Find $x, y \in [0, 2]$ with $|y - x| = 1$ and $f(x) = f(y)$.

1. Compare the values separated by one unit.

For $t \in [0, 1]$, both t and $t + 1$ belong to the domain $[0, 2]$. Define

$$h(t) = f(t + 1) - f(t).$$

Translation $t \mapsto t + 1$ is continuous, so composition makes $t \mapsto f(t + 1)$ continuous on $[0, 1]$. The other term is a restriction of f, so h is continuous.

2. Show that its endpoint values are negatives of one another.

The given equality $f(2) = f(0)$ gives

$$\begin{aligned} h(0) &= f(1) - f(0), \\ h(1) &= f(2) - f(1) = f(0) - f(1) = -h(0). \end{aligned}$$

If $h(0) = 0$, choose $t_0 = 0$. If $h(0) \neq 0$, these two endpoint values have opposite signs. The intermediate value theorem then supplies $t_0 \in (0, 1)$ with $h(t_0) = 0$. Thus in every case there is $t_0 \in [0, 1]$ with $h(t_0) = 0$.

3. Translate the zero back into the requested pair.

Take $x = t_0$ and $y = t_0 + 1$. Then $x \in [0, 1]$ and $y \in [1, 2]$, so both belong to $[0, 2]$. Moreover,

$$|y - x| = |(t_0 + 1) - t_0| = 1, \qquad f(y) - f(x) = h(t_0) = 0.$$

Conclusion. The constructed pair has precisely the required distance and equal function values.

18.11 Exercise 18.11: The decreasing versions of the inverse-function theorems

What we must show. Prove (a) that negating a strictly increasing function makes it strictly decreasing, and (b) state and prove the strictly decreasing versions of Theorems 18.4 and 18.5.

(a) Negation reverses each strict inequality.

If f is strictly increasing on I, then $x_1 < x_2$ in I implies $f(x_1) < f(x_2)$. Multiplication by -1 gives $-f(x_1) > -f(x_2)$. This is exactly the definition of strict decrease of $-f$ on I. The same reasoning shows that negating a strictly decreasing function makes it strictly increasing.

(b), the decreasing version of Theorem 18.5.

Statement. If g is strictly decreasing on an interval J and its image $g(J) = I$ is an interval, then g is continuous on J.

Proof. Let $h = -g$. Part (a), with the inequalities reversed, makes h strictly increasing. Its image is

$$h(J) = -I = \{-y : y \in I\}.$$

Reflection preserves the interval property. Indeed, if $u < v$ belong to $-I$ and $u < w < v$, then $-v < -w < -u$. The endpoints $-v, -u$ belong to I, so the interval property gives $-w \in I$, or $w \in -I$. Thus $-I$ is an interval, with any included endpoints and any infinite endpoints reflected in the usual way.

The increasing version of Theorem 18.5 applies to h: its domain is the interval J, it is strictly increasing, and its image $-I$ is an interval. Therefore h is continuous. Negation is continuous, so $g = -h$ is continuous by Theorem 17.3. Explicitly, the relevant errors are equal:

$$|g(x) - g(x_0)| = |-h(x) + h(x_0)| = |h(x) - h(x_0)|\,.$$

Thus the very same δ that works for h at x_0 works for g. This proves the decreasing version of Theorem 18.5.

(b), the decreasing version of Theorem 18.4.

Statement. If f is continuous and strictly decreasing on an interval I, then its image $J = f(I)$ is an interval, its inverse is a function $f^{-1} : J \to I$, and that inverse is continuous and strictly decreasing on J.

Proof. The intermediate value theorem and Corollary 18.3 make $J = f(I)$ an interval. Strict decrease makes f one-to-one: distinct inputs can be ordered, and that order gives distinct outputs. Thus every $u \in J$ has exactly one preimage in I, which defines $f^{-1}(u)$.

Take $u < v$ in J, and set $x = f^{-1}(u)$ and $y = f^{-1}(v)$. If $x = y$, then $u = f(x) = f(y) = v$, a contradiction. If $x < y$, strict decrease gives $u = f(x) > f(y) = v$, also a contradiction. Hence $x > y$, which means $f^{-1}(u) > f^{-1}(v)$. The inverse is strictly decreasing.

Its domain J is an interval and its image is exactly I, also an interval. The decreasing version of Theorem 18.5 just proved therefore makes f^{-1} continuous on J.

No openness of either image interval has been assumed. Continuity at an included endpoint is relative to J; the definition tests only $u \in J$. If the domain is a singleton, strict monotonicity is vacuous and the singleton inverse is continuous directly, so the statements still hold.

Conclusion. Both decreasing theorems have the same conclusions as their increasing counterparts, with strict decrease replacing strict increase. Reflection and the interval-image condition supply the continuity argument.

18.12 Exercise 18.12: The intermediate value property without continuity

What we must show. For

$$f(x) = \begin{cases} \sin(1/x), & x \neq 0, \\ 0, & x = 0, \end{cases}$$

show (a) discontinuity at zero and (b) the intermediate value property on all of $\mathbb{R}$.

(a) Give an input sequence whose outputs do not approach $f(0)$.

Let

$$u_n = \frac{1}{2\pi n + \pi/2} \qquad (n \in \mathbb{N}).$$

The positive denominators grow without bound, so $u_n \to 0$. But periodicity of sine gives

$$f(u_n) = \sin(2\pi n + \pi/2) = 1 \qquad \text{for every } n,$$

whereas $f(0) = 0$. Thus $f(u_n)$ does not tend to $f(0)$, contradicting the sequential condition for continuity. This is the same discontinuity established in Exercise 17.10(b).

(b), preparation: state exactly what must be checked.

Fix any $a < b$, and let y lie strictly between $f(a)$ and $f(b)$. Every value of f belongs to $[-1, 1]$, so the strict intermediate value y satisfies $-1 < y < 1$. We must find $x \in (a, b)$ with $f(x) = y$.

(b), intervals that stay away from zero.

If $0 \notin [a, b]$, the reciprocal function $1/x$ is continuous on $[a, b]$, and composition with sine makes f continuous there. The ordinary intermediate value theorem then supplies the required interior point.

(b), intervals containing zero and some positive points.

Suppose $0 \in [a, b]$ and $b > 0$. For $n \in \mathbb{N}$, put

$$u_n = \frac{1}{2\pi n + \pi/2}, \qquad v_n = \frac{1}{2\pi n + 3\pi/2}.$$

Both sequences tend to zero, and $0 < v_n < u_n$. Choose n large enough that $u_n < b$. Since $a \leq 0$, the entire interval $[v_n, u_n]$ lies in (a, b) and excludes zero. On its endpoints,

$$f(v_n) = \sin(2\pi n + 3\pi/2) = -1, \qquad f(u_n) = 1.$$

The function is continuous on $[v_n, u_n]$ by the composition argument. Since $-1 < y < 1$, the intermediate value theorem gives $x \in (v_n, u_n) \subseteq (a, b)$ with $f(x) = y$.

(b), the remaining intervals end at zero from the left.

If $0 \in [a, b]$ but $b \leq 0$, then $b = 0$ and $a < 0$. Choose n large enough that $u_n < -a$. Then

$$a < -u_n < -v_n < 0 = b.$$

For nonzero inputs f is odd, because $f(-t) = \sin(-1/t) = -\sin(1/t) = -f(t)$. Therefore $f(-u_n) = -1$ and $f(-v_n) = 1$. The function is continuous on the zero-free interval $[-u_n, -v_n]$. The intermediate value theorem again gives an interior point with value y, and that point lies in (a, b).

These cases cover every possible position of $[a, b]$. When the endpoint values are equal there is no strictly intermediate y to check, so that case imposes no additional condition.

Conclusion. The function is discontinuous at zero but has the intermediate value property on $\mathbb{R}$. Continuity implies that property, while the property alone does not imply continuity.